and choose three courses from the following list:

EENG509	SPARSE SIGNAL PROCESSING	3.0
EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	3.0
EENG511	CONVEX OPTIMIZATION AND ITS ENGINEERING APPLICATIONS	3.0
EENG514	DATA SCIENCE FOR ELECTRICAL ENGINEERING	3.0
EENG517	THEORY AND DESIGN OF ADVANCED CONTROL SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0
EENG589	DESIGN AND CONTROL OF WIND ENERGY SYSTEMS	3.0

Required Core: Power and Energy Systems Track

All students must choose four courses from the following list:

EENG570	ADVANCED HIGH POWER ELECTRONICS	3.0
EENG573	ELECTRIC POWER QUALITY	3.0
EENG577	ADVANCED ELECTRICAL MACHINE DYNAMICS FOR SMART-GRID SYSTEMS	
EENG581	POWER SYSTEM OPERATION AND MANAGEMENT	3.0
EENG584	POWER SYSTEM RISK MANAGEMENT	3.0
EENG585	AI FOR POWER AND RENEWABLE ENERGY SYSTEMS	
EENG587	POWER SYSTEMS PROTECTION AND RELAYING	3.0
EENG588	POWER SYSTEM ECONOMICS AND ELECTRICITY MARKETS	3.0

Professional Online Masters in Electrical Engineering

The professional master's degree is designed to train and target recent graduates or mid-career professionals with a B.S. in electrical engineering or a related field in physics or applied sciences. The program is composed of 3 stackable certificates plus a required graduate-level mathematics course (MEGN502). To complete the professional master's degree the student must complete 30 credits as outlined below.

Students may also register for the professional masters at the outset and take a mixture of courses (from the tracks) in any order. Should the student elect to register for the full masters, then certificates will not be awarded on completion of the full degree. For these students, we refer to the certificates as 'tracks' as no certificates will be awarded.

Certificate 1 / Track 1: Information and System Sciences – students must complete 9 hours of coursework as follows:

EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	
EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0

Certificate 2 / Track 2 : Microwave Engineering – students must complete 9 hours of coursework as follows:

EENG530	PASSIVE RF & MICROWAVE DEVICES
EENG532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING
EENG533	ACTIVE RF & MICROWAVE DEVICES

Certificate 3 / Track 3: Power & Energy Systems – students must complete 9 hours of coursework from among the following courses:

EENG570	ADVANCED HIGH POWER ELECTRONICS	3.0
EENG577	ADVANCED ELECTRICAL MACHINE DYNAMICS FOR SMART-GRID SYSTEMS	
EENG585	AI FOR POWER AND RENEWABLE ENERGY SYSTEMS	3.0
EENG588	POWER SYSTEM ECONOMICS AND ELECTRICITY MARKETS	3.0

Graduate certificate in information and system sciences

The post-baccalaureate certificate program in Information and System Sciences is targeted to train recent graduates or mid-career professionals with a basic knowledge of modeling and extracting information from signals, systems, or data sets.

To earn the post-baccalaureate certificate program in Information and System Sciences, students must complete 9 hours of coursework as follows:

Required Courses:

EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	3.0
EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0

While all courses are to be delivered online, students that are interested in attending an on-campus class may substitute one of the above courses with one of the Information and Systems Sciences track electives as listed in our Graduate Catalog.

Graduate certificate in microwave engineering

The online certificate program in Microwave Engineering is targeted to train recent graduates or mid-career professionals with a B.S. in electrical engineering or a related field in physics or applied sciences with a basic knowledge of electromagnetic theory, specifically for handling the challenges and demands of modern microwave systems.

To complete the online certificate program in Microwave Engineering, students must complete 9 hours of coursework as follows:

Required Courses:

EENG530	PASSIVE RF & MICROWAVE DEVICES
EENG532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING
EENG533	ACTIVE RF & MICROWAVE DEVICES

While all courses are to be delivered online, students that are interested in attending an on-campus class may substitute one of the above courses with one of the Antennas & Wireless electives as listed in our Graduate Catalog.

graduate certificate in power and energy systems

The online certificate program in Power and Energy Systems is targeted to train recent graduates or mid-career professionals with a B.S. in electrical engineering or a related field in physics or applied sciences with a basic knowledge of power systems and machines.

To complete the online certificate program in Power and Energy Systems, students must complete 9 hours of coursework as follows:

Required Courses:

EENG570	ADVANCED HIGH POWER ELECTRONICS	3.0
EENG577	ADVANCED ELECTRICAL MACHINE DYNAMICS FOR SMART-GRID SYSTEMS	3.0
EENG585	AI FOR POWER AND RENEWABLE ENERGY SYSTEMS	

While all courses are to be delivered online, students that are interested in attending an on-campus class may substitute one of the above courses with one of the Power and Energy Systems electives as listed in our Graduate Catalog.

Courses

EENG500. ELECTRICAL ENGINEERING SEMINAR. 0.0 Semester Hrs.
This zero-credit graduate course builds on the EE department seminars in the colloquium series, which consist of presentations delivered by external or internal invited speakers on topics broadly related to electrical engineering. The seminar is mandatory for all graduate students (MS and Ph.D.). The students would need to enroll in the course every semester. Any student who cannot take the course for valid reasons should notify their adviser, who will then make a request to the EE graduate committee for a waiver. These requests could be for the duration of one semester or longer. The course will be graded as PRG/PRU based on student attendance at the department seminars in the colloquium series - the student has to attend at least two thirds of all the seminars each semester in order to get a PRG grade.
Course Learning Outcomes

- Graduates will demonstrate the ability to conduct directed research.
- Graduates will demonstrate oral and written communication skills.

EENG507. INTRODUCTION TO COMPUTER VISION. 3.0 Semester Hrs.
Equivalent with CSCI507,CSCI512,EENG512,
Computer vision is the process of using computers to acquire images, transform images, and extract symbolic descriptions from images. This course provides an introduction to this field, covering topics in image formation, feature extraction, location estimation, and object recognition. Design ability and hands-on projects will be emphasized, using popular software tools. The course will be of interest both to those who want to learn more about the subject and to those who just want to use computer imaging techniques.
Course Learning Outcomes

- 1. Be able to analyze and predict the behavior of image formation, transformation, and recognition algorithms.
- 2. Be able to design, develop, and evaluate algorithms for specific applications.
- 3. Be able to use software tools to implement computer vision algorithms.
- 4. Communicate (in oral and written form) methods and results to a technical audience.

EENG509. SPARSE SIGNAL PROCESSING. 3.0 Semester Hrs.
This course presents a mathematical tour of sparse signal representations and their applications in modern signal processing. The classical Fourier transform and traditional digital signal processing techniques are extended to enable various types of computational harmonic analysis. Topics covered include time-frequency and wavelet analysis, filter banks, nonlinear approximation of functions, compression, inverse problems, compressive sensing, and connections with machine learning. Prerequisites: EENG411 and EENG515 or instructor consent.
Course Learning Outcomes

- Students will develop the link between the Fourier, time-frequency, and wavelet transforms.
- Compute and analyze linear and nonlinear approximations of functions.
- Students will be able to use sparse signal representations for solving signal restoration and inverse problems.

EENG510. ADVANCED DIGITAL SIGNAL PROCESSING. 3.0 Semester Hrs.
Equivalent with CSCI510,EGGN510,
This course covers mathematical and engineering aspects of digital signal processing (DSP). An emphasis is placed on the various possible representations for discrete-time signals and systems (in the time, z-, and frequency domains) and how those representations can facilitate the identification of signal properties, the design of digital filters, and the sampling of continuous-time signals. Deterministic and random signal and noise models are discussed, as are methods for noise removal and power spectrum estimation. Additional topics include multi-rate signal processing and spectral analysis using the discrete Fourier transform. The course will be useful to all students who are concerned with information bearing signals and signal processing in a wide variety of application settings, including sensing, instrumentation, control, communications, signal interpretation and diagnostics, and imaging. Prerequisite: EENG310, EENG 311, EENG 391; or consent of instructor.
Course Learning Outcomes

- Design digital filters to particular specifications (passband, cutoff, order, etc.)
- Use digital filters to process analog signals (appreciating the roles of sampling, aliasing, digital filtering, sample rate conversion, and interpolation).
- Analyze the frequency spectrum of digital and sampled analog signals using windowing and the discrete Fourier transform (DFT).
- Estimate the power spectrum of random signals.
- Implement digital signal processing techniques in computational software.

EENG511. CONVEX OPTIMIZATION AND ITS ENGINEERING APPLICATIONS. 3.0 Semester Hrs.
The course focuses on recognizing and solving convex optimization problems that arise in applications in various engineering fields. Covered topics include basic convex analysis, conic programming, duality theory, unconstrained optimization, and constrained optimization. The application